

Git Clone, Pull and Fetch Assignment

1. Assignment Objective

This assignment is focused on downloading an existing GitHub repository, opening it locally, making changes, pushing changes back to GitHub, and then bringing GitHub changes back into the local system.

By the end of this assignment, students should understand:

```
git clone = first-time download from GitHub to computer
git pull = get latest changes from GitHub to local project
git fetch = check/download latest information from remote without directly updating working files
git push = upload committed local changes to GitHub
cd = move into a folder
code . = open current folder in VS Code
```

Main memory line:

```
Clone once. Pull many times. Push after commit.
```

2. Tools Required

Students need:

- Git installed
- GitHub account
- VS Code installed
- Terminal or Git Bash
- Internet connection

- One GitHub repository already available

Recommended terminal:

- Windows: Git Bash, PowerShell, or VS Code terminal
 - macOS: Terminal or VS Code terminal
 - Linux: Terminal or VS Code terminal
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3. Quick Recap Before Assignment

Push

Push sends committed local changes to GitHub.

Computer → GitHub

Command:

```
git push
```

Clone

Clone downloads a GitHub repository to the computer for the first time.

GitHub → Computer

Command:

```
git clone <repo-url>
```

Pull

Pull brings the latest GitHub changes into an already existing local project.

GitHub → Existing Local Project

Command:

```
git pull
```

Fetch

Fetch contacts GitHub and gets information about latest remote changes.

GitHub information → Local Git database

Command:

```
git fetch
```

Important explanation:

```
git pull = git fetch + git merge
```

For beginners, explain it simply as:

```
git pull first checks GitHub for new changes, then brings those changes into your current local project.
```

4. Assignment Task

Task Name

Clone a GitHub repository, update it locally, push it, update it on GitHub, and pull the latest changes locally.

Repository Requirement

Use an existing GitHub repository that already contains:

```
index.html  
README.md
```

Example repository name:

```
git-github-assignment
```

5. Step 1: Copy GitHub Repository URL

Open your GitHub repository in browser.

Click:

```
Code → Local → HTTPS → Copy URL
```

The URL will look like this:

```
https://github.com/username/git-github-assignment.git
```

Use the HTTPS URL for this assignment.

6. Step 2: Clone the Repository

Open terminal in a normal folder such as Desktop or Documents.

Run:

```
git clone <repo-url>
```

Example:

```
git clone https://github.com/username/git-github-assignment.git
```

This command creates a folder automatically.

Example:

```
git-github-assignment/
```

7. Step 3: Clone with a Custom Folder Name

You can also clone the same repository into a custom folder name.

Syntax:

```
git clone <repo-url> <folder-name>
```

Example:

```
git clone https://github.com/username/git-github-assignment.git my-cloned-project
```

This creates a folder named:

```
my-cloned-project/
```

Use this when you want a different local folder name.

8. Step 4: Move Into the Project Folder

After cloning, enter the cloned project folder.

```
cd <folder-name>
```

Example:

```
cd git-github-assignment
```

If you used a custom folder name:

```
cd my-cloned-project
```

Check current Git status:

```
git status
```

Important:

Do not run `git init` after `git clone`.
A cloned project is already a Git repository.

9. Step 5: Open Project in VS Code

After entering the folder, open it in VS Code:

```
code .
```

Meaning:

Open the current folder in VS Code.

If `code .` does not work, manually open VS Code and choose:

File → Open Folder → select cloned project folder

10. Step 6: Modify `index.html`

Open `index.html` and add one new line before `</body>`.

Add this:

```
<p>This line was added after cloning the repository.</p>
```

Save the file.

11. Step 7: Commit and Push Local Change

Check status:

```
git status
```

Add the changed file:

```
git add .
```

Commit the change:

```
git commit -m "updated index file after clone"
```

Push to GitHub:

```
git push
```

Refresh GitHub and verify the new line is visible in `index.html`.

12. Step 8: Edit File Directly on GitHub

Now make one change directly on GitHub.

Steps:

1. Go to the GitHub repository.

2. Open `index.html`.
3. Click the edit button.
4. Add this line before `</body>`:

```
<p>This line was added directly from GitHub.</p>
```

1. Commit the change on GitHub.

Now GitHub has a new change that is not yet available in your local folder.

13. Step 9: Use `git fetch`

Come back to terminal inside the cloned project folder.

Run:

```
git fetch
```

Simple meaning:

```
git fetch checks GitHub and downloads information about new remote changes.
```

For beginners, remember:

```
fetch checks remote changes, but pull brings them into your current project files.
```

14. Step 10: Use `git pull`

Now bring the GitHub change into your local project:

```
git pull
```

After this, open `index.html` locally.

You should see the line added from GitHub:

```
<p>This line was added directly from GitHub.</p>
```

Important explanation:

```
git pull = git fetch + git merge
```

For now, understand it as:

```
git pull brings the latest GitHub changes into your local project.
```

15. Step 11: Pull From Origin Main

Sometimes we write the remote and branch name explicitly:

```
git pull origin main
```

Meaning:

```
Pull latest changes from the main branch of the origin remote.
```

For today:

```
origin = GitHub remote name  
main = main code line
```

Use this when your trainer asks you to specify the remote and branch clearly.

16. Commands to Practice

Students should practice these commands in order:

```
git clone <repo-url>
git clone <repo-url> <folder-name>
cd <folder-name>
code .
git status
git add .
git commit -m "updated index file after clone"
git push
git fetch
git pull
git pull origin main
```

17. Final Self-Check

At the end of the assignment, students should be able to answer:

- Which command downloads a repository for the first time?
- Which command enters a folder?
- Which command opens the current folder in VS Code?
- Which command uploads local commits to GitHub?
- Which command checks remote updates?
- Which command brings latest GitHub changes into local project?
- What is the difference between `git fetch` and `git pull`?
- Why should we not run `git init` after `git clone`?

Final summary:

```
git clone = first-time download
git fetch = check/download remote information
git pull = fetch + merge
git push = upload committed changes
cd = enter folder
code . = open current folder in VS Code
```